From User Stories to Source Code: Investigating the impact of requirement quality on LLM generated code

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Objective

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- Investigate how the quality and type of user stories influence code generation by LLMs, analyzing the fulfillment of functional requirements and the presence of code smells and unnecessary

Research Questions

- **RQ1: what is the impact of the user stories quality on the presence of code smells in a code ?**
  - Code metrics: number of the code smells detected ,how (duplicated code, god class, etc..)
  - Define four or more code smells specifics
- **RQ2: Can the generated code correctly meet all the requirements of the user stories ?**
  - I want to evaluate if one generated code have success
  - I need to think if this evaluate will be used IA or peoples
  - I will used fragmente of the code to developer evaluate same code
- **RQ3: Does the type of story influence the presence of unnecessary complexity in the code?**
  - I want to evaluate which code is better generated

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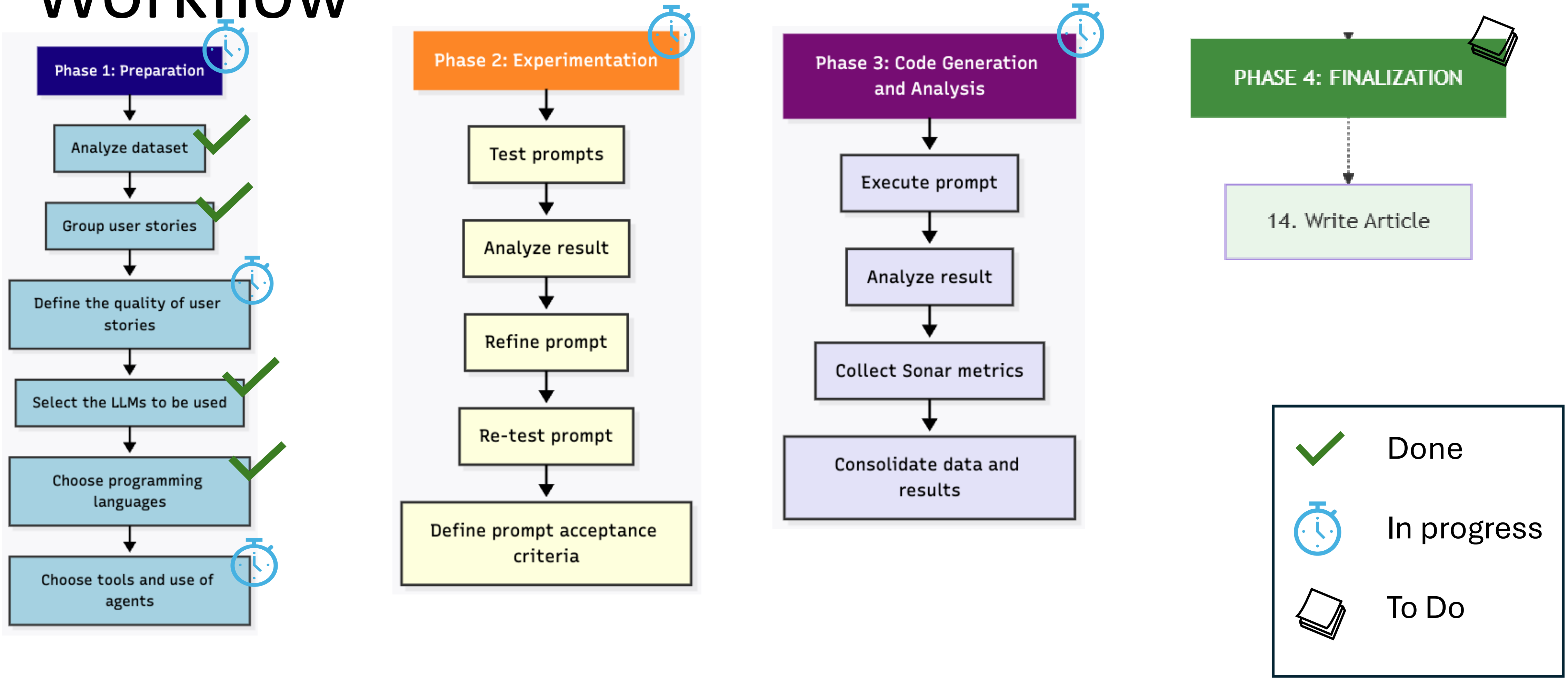
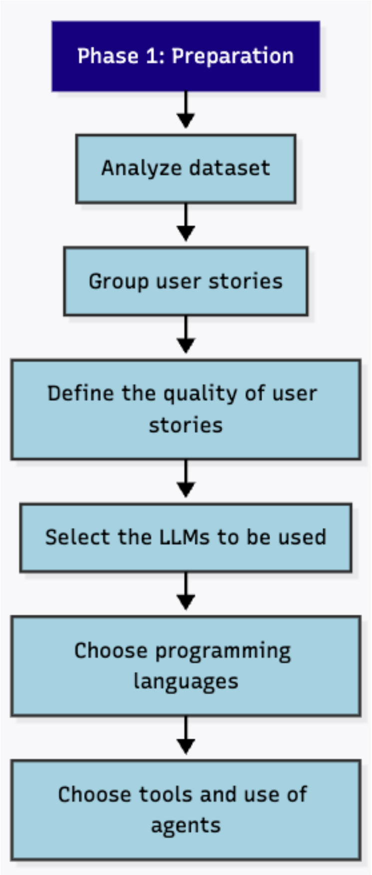
Workflow



Phase 1: Preparation
Analyze dataset
Group user stories
Define the quality of user stories
Select the LLMs to be used
Choose programming languages
Choose tools and use of agents
Phase 2: Experimentation
Test prompts
Analyze result
Refine prompt
Re-test prompt
Define prompt acceptance criteria
Phase 3: Code Generation and Analysis
Execute prompt
Analyze result
Collect Sonar metrics
Consolidate data and results
PHASE 4: FINALIZATION
14. Write Article
Done
In progress
To Do

Phase 1: Preparation

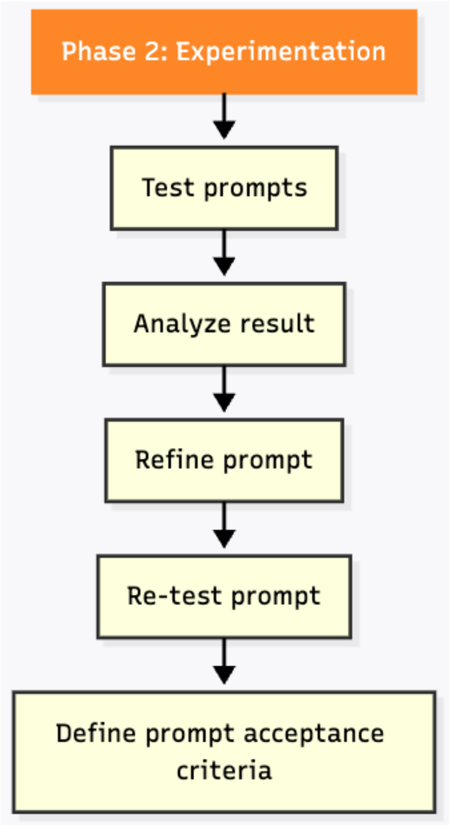


- Dataset: Green Market and TechFix (Description, main flow, alternative flow)
- Many stories with similar settings.
- To reliably define what constitutes a good user story.

- Chosen programming languages
  - Java
  - Python
  - Javascript
- Chosen LLMs
  - gpt-5.3-codex.high
  - clude sonnet 4.6
  - ~~gemini 3 flash preview~~
  - Gemini 2.5 pro

- Github copilot

Phase 2: Experimentation



- Test 1:
- Zero Shot short with user stories

- Next tests
  - Try out new types of prompts.
    - Few-shot
    - Chain-of-Thought

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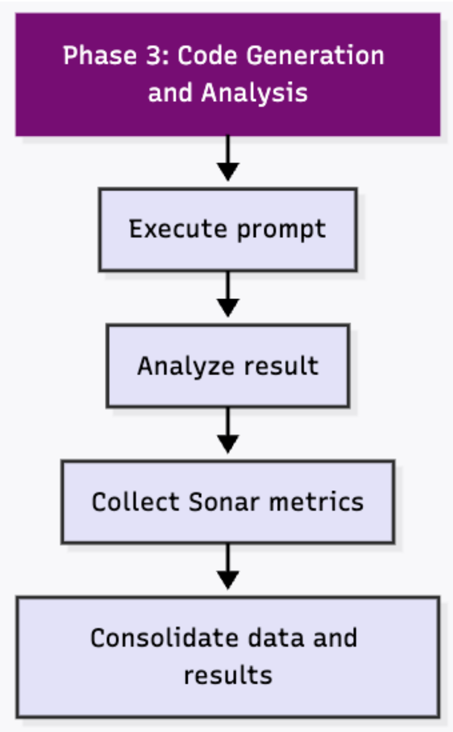
Test 1

- Zero Shot short

"Act as a Java specialist developer and implement the set of user stories contained in the file. "

Phase 3: Code generation and analysis



LLM	Language	QTd Code smell	LOC(Line of code)	Bugs by sonar	Cyclomatic Complexity	Cognitive Complexity
gpt-5.3-codex.medium	Java	0	403	0	76	27
gemini 3 flash preview	java	23	385	0	85	23
clude sonnet 4.6	java	0	851	0	89	28
gpt-5.3-codex.high	python	0	440	0	42	27

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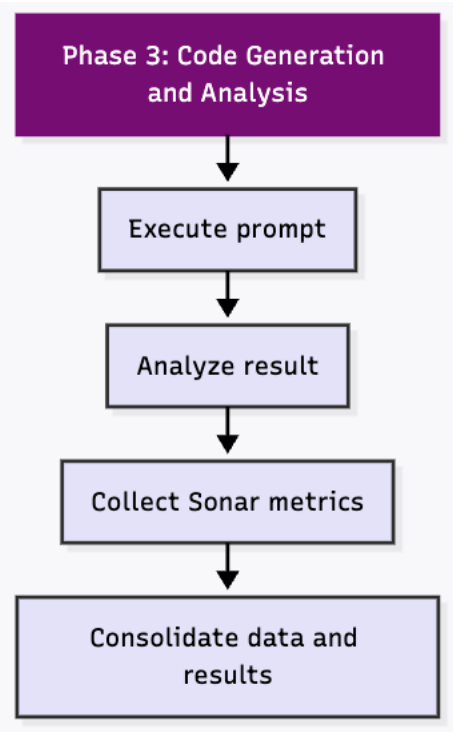
Phase 3: Next steps



- Generate code for other languages

Next steps

- Generate code for other languages;.
- Test approaches such as:
- SDD (Specification driven develop),
- Analyze agent behavior and check if there is a difference between using an API or integrated into the IDE.

Thank you!

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